Machine Learning Identification of Ionization Cones

William Olson[1], John Schneider[1], Lulu Zhang[1], Chris Packham[12]
[1] The University of Texas at San Antonio, Texas, USA
[2] National Astronomical Observatory of Japan, Tokyo, Japan

william.olson@my.utsa.edu
+1 (210) 274-3958

Introduction

Ionization cones are observational signatures of active galactic nuclei (AGN), revealing information about nuclear geometry, obscuration, and feedback.

Determining these structures is challenging due to:

- Projection effects
- Host galaxy contamination
- irregular morphologies

Current approaches often rely on manual inspection or model-dependent fitting, limiting scalability across large datasets.

Objective

Develop a machine learning framework to identify ionization cones and recover their morphology from imaging data.

Pipeline Overview

We formulate ionization cone recovery as an image-to-segmentation problem.

Input emission-line images are processed through a convolutional neural network to produce pixel-wise masks of ionized structures, to compare geometry.

Pipeline:

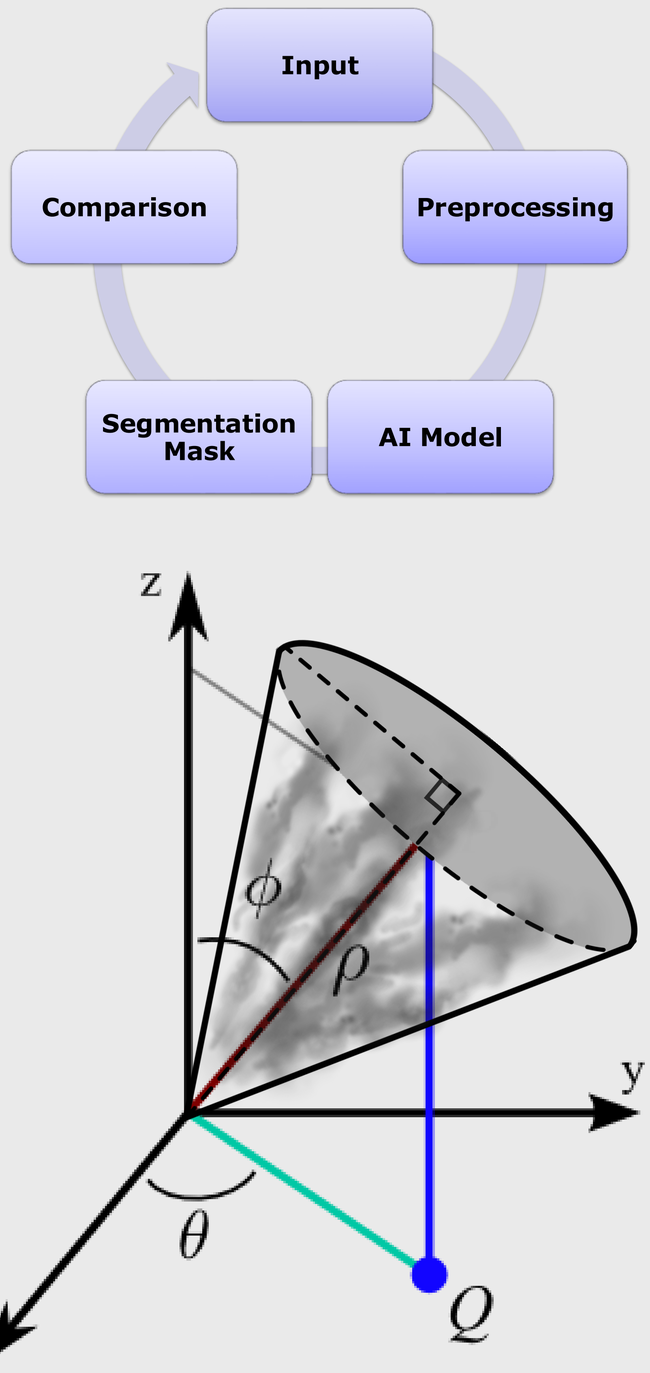


Figure 1. Input Geometry
- 3D model of an AGN ionization cone centered on a black hole (origin).
- The cone geometry is controlled by the parameters (θ,ϕ).
- 3D emission is projected into a 2D brightness map used as input for the model.

Methodology

We use a U-Net convolutional neural network for semantic segmentation of ionized emission.

U-Net Architecture

- Encoder extracts multi-scale spatial features
- Decoder reconstructs emission morphology
- Skip connections preserve fine spatial structure

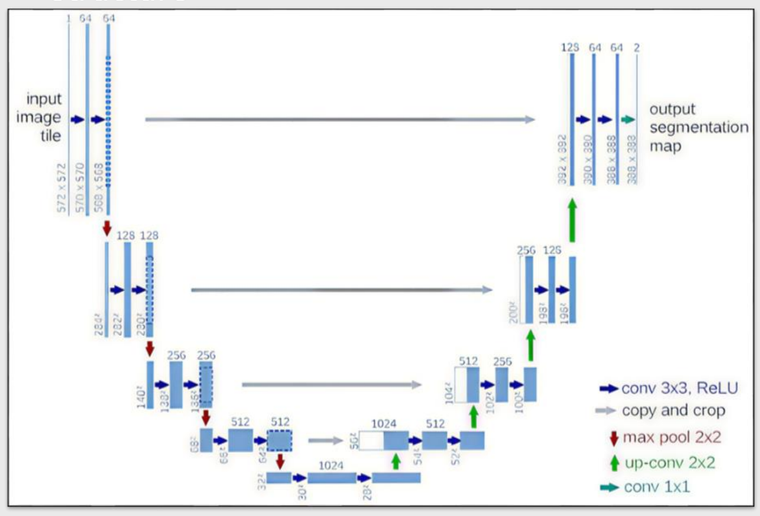


Figure 2. The U-Net Architecture (Credit Ronneberger et. al 2015)

This enables accurate segmentation of diffuse, extended emission structures.

Synthetic Modeling

Synthetic training data is designed to reproduce key properties of AGN ionization cones:

- Biconic geometry
- Opening angle variation
- Orientation and inclination effects
- Clumpy gas structure with radial flux falloff
- Cone asymmetry
- Observational noise and host galaxy contamination

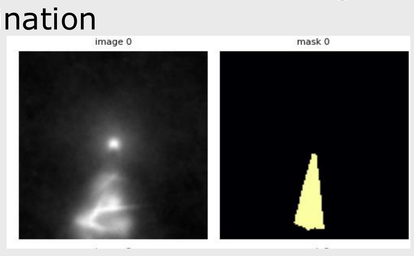


Figure 3. Example of a Synthetic Input
Added gaussian and fragmentation to simulate diffuseness.

Geometric Extraction

The segmentation mask is treated as a projection of a biconical outflow.

- Cone axis inferred from emission morphology
- Opening angle computed from angular extent of segmented region

Simulated Performance

Table 1. Synthetic Validation

Metric	
Mean IoU (Average Overlap)	0.647 ± 0.199
Mean ϕ (Position Angle) Error	$3.7° \pm 2.7°$
Mean ρ (Opening Angle) Error	$13.1° \pm 7.1°$

Metrics for a 1000 training synthetic input training sample support the stability and reliability for U-Net cone segmentation mask.

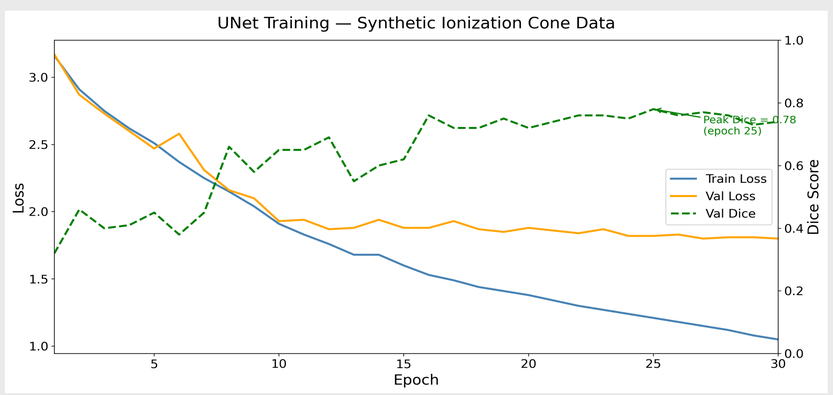


Figure 3. The Loss function graph.

Real Prediction

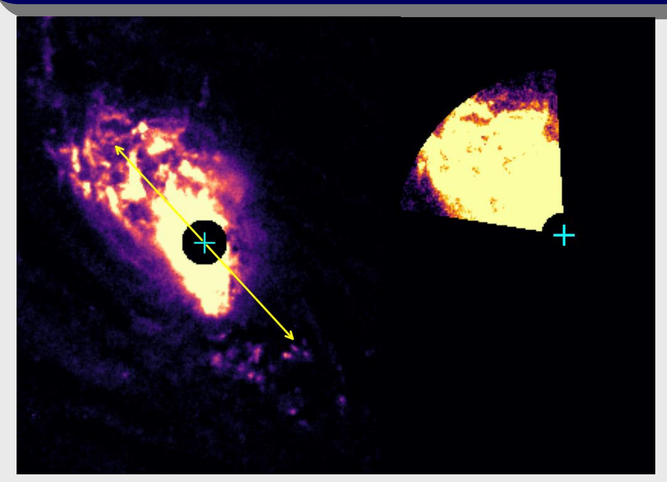


Figure 4. NGC 1068 Adapted from HST F502N MAST
Pictured left, the continuum subtracted ionized gas of NGC 1068 fed to the model.
Pictured right, the real calculated conical area. This is our ground truth.

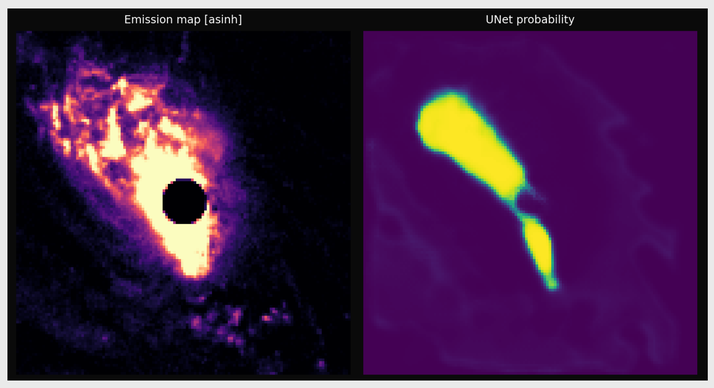


Figure 5. Trained U-Net prediction on NGC-1068 processed ionized gas.

Recovered parameters vs. reference values:

- ϕ (Position Angle): 136.3 vs. 131.8 (3.4% error)
- ρ (Opening Angle): 33.9 vs. 80.0 (57.6% error)

Conclusions

- Framework Success: U-Net architecture successfully extracts ionization cone morphology from complex, clumpy imaging.
- Proof of Concept: Model shows successful foundational transfer from synthetic training to real observations of NGC 1068.
- Geometric Recovery: Recovered Position Angle for NGC 1068 is within 3.4% of reference values.
- Inclination Sensitivity: Significant error in ρ (57.6%) highlights sensitivity to brightness & synthetic naivity.
- Scalability: Provides a consistent, automated alternative to manual inspection for large-scale AGN surveys.

Future Development

- Sample Expansion: Applying the pipeline to the full GATOS sample and nearby Seyferts.
- Physics Refinement: Greatly improving synthetic modeling of host galaxy extinction and line-of-sight projection.

References

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